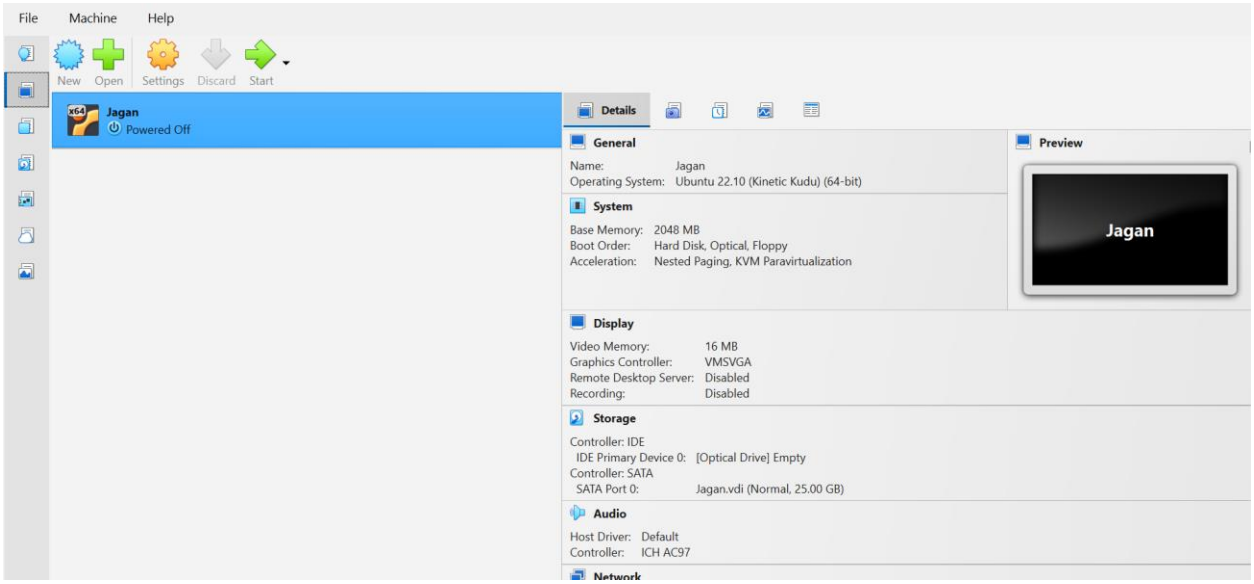
Setup a virtual machine using Ubuntu image.



Install docker on ec2.

```
[ec2-user@ip-10-0-13-233 ~]$ sudo su -
[root@ip-10-0-13-233 ~]# apt update
-bash: apt: command not found
[root@ip-10-0-13-233 ~]# dnf update -y
Amazon Linux 2023 Kernel Livepatch repository                                   920 kB/s |  89 kB     00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-10-0-13-233 ~]# dnf install docker -y
Last metadata expiration check: 0:00:26 ago on Mon Oct 12 11:53:42 2026.
Dependencies resolved.
================================================================================
 Package                  Architecture          Version                  Repository                Size
================================================================================
Installing:
 docker                   x86_64                25.0.16-1.amzn2023.0.4   amazonlinux               46 M
Installing dependencies:
 container-selinux        noarch                4:2.245.0-1.amzn2023     amazonlinux               58 k
 containerd               x86_64                2.2.7-1.amzn2023.0.1     amazonlinux               25 M
```

Create ec2 on docker and install docker

```
[root@ip-10-0-13-233 ~]# systemctl start docker
[root@ip-10-0-13-233 ~]# systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; disabled; preset: disabled)
     Active: active (running) since Mon 2026-10-12 11:54:58 UTC; 5s ago
TriggeredBy: ● docker.socket
```

```
[root@ip-10-0-13-233 ~]# docker --version
Docker version 25.0.14, build 0bab007
```

Pull 5 docker images. (nginx, Apache tomcat, ubuntu, Jenkins, SonarQube)

```
[root@ip-10-0-13-233 ~]# docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
ecc510c1e359: Pull complete
04bdbc7ea6b5: Pull complete
807ab930f3a0: Pull complete
1efc53192927: Pull complete
947d22046304: Pull complete
8020170dc65f: Pull complete
f7e1db57b58b: Pull complete
Digest: sha256:f9ea18bfa4fad859e1ed38259d711da7ccad2c3516e875cec3351a57c859f571
Status: Downloaded newer image for nginx:latest
```

Nginx is done

```
[root@ip-10-0-13-233 ~]# docker pull tomcat
Using default tag: latest
latest: Pulling from library/tomcat
4e07a0f12b2c: Pull complete
06ad70e463aa: Pull complete
5dec61325458: Pull complete
f39241fdc590: Pull complete
4f4fb700ef54: Pull complete
130739ca07b1: Pull complete
6d27111e20e8: Pull complete
a61c6f59ba30: Pull complete
Digest: sha256:bb7c0c78078841047e2d475acce833ef337cb0007ca7c7e4c9ff9058d9019801
Status: Downloaded newer image for tomcat:latest
docker.io/library/tomcat:latest
```

Tomcat installed

```
[root@ip-10-0-13-233 ~]# docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
4e07a0f12b2c: Already exists
06ad70e463aa: Already exists
Digest: sha256:f144425ff09be612d6d9ad965196e9cdc23dae1f42110a8a11a3e9a8198759f7
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
```

Ubuntu installed

```
[root@ip-10-0-13-233 ~]# docker pull jenkins/jenkins:lts
lts: Pulling from jenkins/jenkins
6eefb2f5d3e9: Pull complete
408777b27aa9: Pull complete
810008851f3a: Pull complete
d89089c8a5a3: Pull complete
701ded2f9b33: Pull complete
968c36a3b2c5: Pull complete
3dd4a99993ab: Pull complete
a20a14499a3e: Pull complete
463a574793f8: Pull complete
f9e31fab50e8: Pull complete
95bc7464aaea: Pull complete
a91427fe0e3f: Pull complete
Digest: sha256:a660310e39ade10631f774bacd5219de767dfd08947ea5c32a739b5e5bb382c1
Status: Downloaded newer image for jenkins/jenkins:lts
docker.io/jenkins/jenkins:lts
```

Jenkins installed

```
[root@ip-10-0-13-233 ~]# docker pull sonarqube
Using default tag: latest
latest: Pulling from library/sonarqube
a60ef8ec184e: Pull complete
96fe3bd0148e: Pull complete
54929e68a7af: Pull complete
4f4fb700ef54: Pull complete
0a66c41acdcb: Pull complete
065adb12f4f5: Pull complete
03217b7e6be2: Pull complete
Digest: sha256:c0f1160bccfa435db4168c2d7df69af3c3ea7bfeb87395613014048fb33a68c1
Status: Downloaded newer image for sonarqube:latest
docker.io/library/sonarqube:latest
```

Sonarqube installed

```
[root@ip-10-0-13-233 ~]# docker images
REPOSITORY        TAG       IMAGE ID       CREATED       SIZE
nginx             latest    57df7715e516   6 days ago    163MB
tomcat            latest    04dd9f6a2a84   6 days ago    444MB
sonarqube         latest    01676bb7354b   10 days ago   1.44GB
jenkins/jenkins   lts       82becee63645   12 days ago   445MB
ubuntu            latest    6e92c4fd8dfd   2 weeks ago   101MB
```

Run nginx container and expose on port 81.

```
[root@ip-10-0-13-233 ~]# docker run -d --name nginx-container -p 81:80 nginx
3da508b14d526d40a809f87a1cd2671ffdceb6338bd6c1020fe9d2906fd487ce
```

What it means:

-d — runs the container in the background.

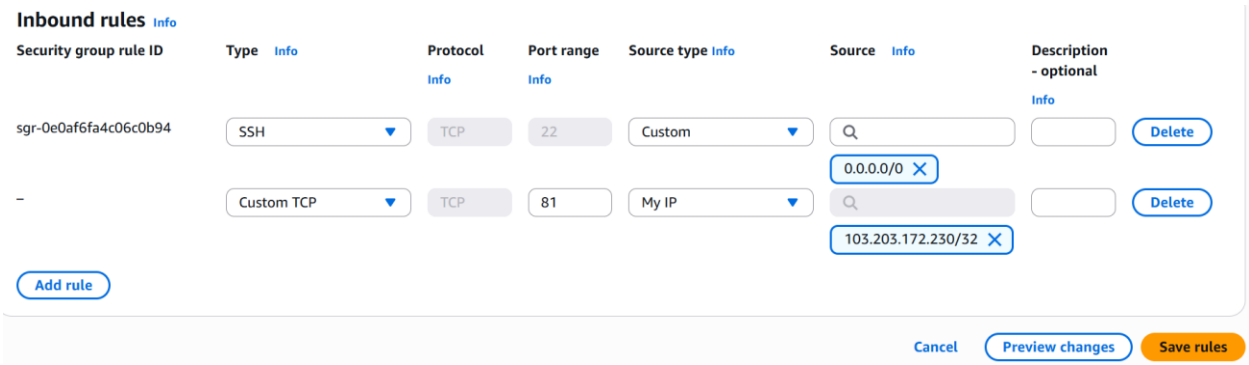
--name nginx-container — names your container.

-p 81:80 — maps EC2 port 81 to Nginx container port 80.

nginx — uses the Nginx image you downloaded.

```
[root@ip-10-0-13-233 ~]# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED              STATUS          PORTS                                   NAMES
3da508b14d52   nginx     "/docker-entrypoint.…"   About a minute ago   Up 59 seconds   0.0.0.0:81->80/tcp, :::81->80/tcp   nginx-container
```

Docker ps to verify



Add inbound rules allowing 81



Nginx is working

```
[root@ip-10-0-13-233 ~]# docker images
REPOSITORY        TAG       IMAGE ID       CREATED       SIZE
nginx             latest    57df7715e516   6 days ago    163MB
tomcat            latest    04dd9f6a2a84   6 days ago    444MB
sonarqube         latest    01676bb7354b   10 days ago   1.44GB
jenkins/jenkins   lts       82becee63645   12 days ago   445MB
ubuntu            latest    6e92c4fd8dfd   2 weeks ago   101MB
```

Delete the Apache tomcat image from local.

```
[root@ip-10-0-13-233 ~]# docker rmi tomcat
Untagged: tomcat:latest
Untagged: tomcat@sha256:bb7c0c78078841047e2d475acce833ef337cb0007ca7c7e4c9ff9058d9019801
Deleted: sha256:04dd9f6a2a84def11e370ff7e96102021043d0c115cb728926202b8621fb94ca
Deleted: sha256:90b02ae599999b3e0a2e9319aefe2d46249cd3a84b04a4256a73fe4ce66a8136
Deleted: sha256:8b4bcc344f173d148c3ca8a755516b8e526f28fcd6a77dca5e28c7873bfa2e76
Deleted: sha256:e434504b45aa7e0af3face17651b8b92ad976aaed02a1e1d0eba614358138aa3
Deleted: sha256:19280388cd8fa31b351a4cf0daa0e9175be81372ed702b33c7aa5eb5bbce8b4e
Deleted: sha256:aed1f647f35084f548ce1265b300b650aa150eaa0bfcf2e99f685c17bc1ab5de
Deleted: sha256:d48beed04406c077ac1cb038fd819d5650881bc4c1aafc17156635551e4503e6
Deleted: sha256:8ee0776a2d1bb912f27e879c3140cc88f29eee8f8b6e093f58205bc190b90443
Deleted: sha256:9aa61dc7dab0b58091261860aefe01de30bcf24fb226890fd1cf936230e7588f
```

Use {docker rmi tomcat} to delete tomcat image

```
[root@ip-10-0-13-233 ~]# docker images
REPOSITORY        TAG       IMAGE ID       CREATED       SIZE
nginx             latest    57df7715e516   6 days ago    163MB
sonarqube         latest    01676bb7354b   10 days ago   1.44GB
jenkins/jenkins   lts       82becee63645   12 days ago   445MB
ubuntu            latest    6e92c4fd8dfd   2 weeks ago   101MB
```

```
[root@ip-10-0-13-233 ~]# docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                   NAMES
3da508b14d52   nginx     "/docker-entrypoint.…"   7 minutes ago   Up 7 minutes   0.0.0.0:81->80/tcp, :::81->80/tcp       nginx-container
```

Inspect the Jenkins image, SonarQube image.

```
[root@ip-10-0-13-233 ~]# docker image inspect jenkins/jenkins:lts
[
    {
        "Id": "sha256:82becee636459f5f9d5195b47b76c891b49cb8c6cf0913986b63e0b3e4b3328b",
        "RepoTags": [
            "jenkins/jenkins:lts"
        ],
        "RepoDigests": [
            "jenkins/jenkins@sha256:a660310e39ade10631f774bacd5219de767dfd08947ea5c32a739b5e5bb382c1"
        ],
        "Parent": "",
        "Comment": "buildkit.dockerfile.v0",
        "Created": "2026-09-30T08:56:28.052436328Z",
        "Container": "",
        "ContainerConfig": {
            "Hostname": "",
            "Domainname": "",
            "User": "",
            "AttachStdin": false,
            "AttachStdout": false,
            "AttachStderr": false,
```

docker image inspect jenkins/jenkins:lts to check the details of the jenkins

```
[root@ip-10-0-13-233 ~]# docker image inspect sonarqube:latest
[
    {
        "Id": "sha256:01676bb7354b8b36a5f3d6b47b106acc08c5ca9a24490f38972e38d5832b7448",
        "RepoTags": [
            "sonarqube:latest"
        ],
        "RepoDigests": [
            "sonarqube@sha256:c0f1160bccfa435db4168c2d7df69af3c3ea7bfeb87395613014048fb33a68c1"
        ],
        "Parent": "",
        "Comment": "buildkit.dockerfile.v0",
        "Created": "2026-10-02T02:34:24.202145562Z",
        "Container": "",
        "ContainerConfig": {
            "Hostname": "",
            "Domainname": "",
            "User": "",
            "AttachStdin": false,
            "AttachStdout": false,
```

docker image inspect sonarqube:latest to check the details of sonarqube

```
[root@ip-10-0-13-233 ~]# docker image inspect jenkins/jenkins:lts --format='{{json .Config.ExposedPorts}}'
{"50000/tcp":{},"8080/tcp":{}}
```

docker image inspect jenkins/jenkins:lts --format='{{json .Config.ExposedPorts}}'

This command is used to check the port, jenkins running on 8080

```
[root@ip-10-0-13-233 ~]# docker image inspect sonarqube:latest --format='{{json .Config.ExposedPorts}}'
{"9000/tcp":{}}
```

Sonarqube running on port 9000

Run Jenkins container and run one sample job.

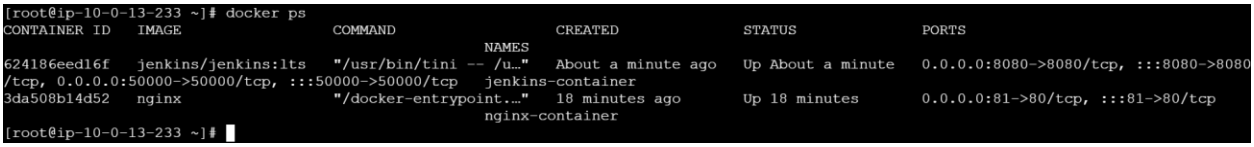
```
[root@ip-10-0-13-233 ~]# docker run -d \
  --name jenkins-container \
  -p 8080:8080 \
  -p 50000:50000 \
  jenkins/jenkins:lts
624186eed16fe0bc6b80a0fb7886f48b542b62ed9c6bae316ab7ab9e2f848282
[root@ip-10-0-13-233 ~]#
```

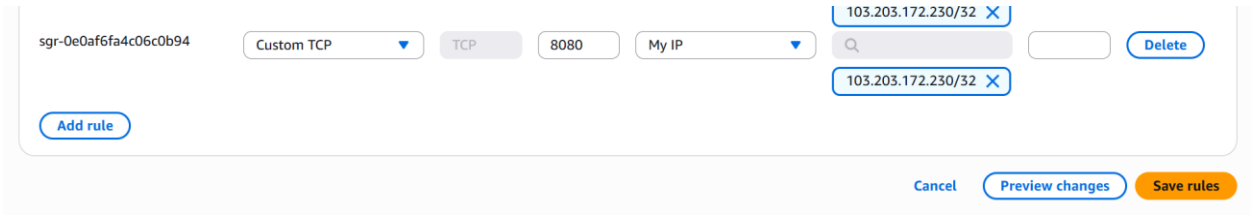
-d — runs in the background.

--name — names the container.

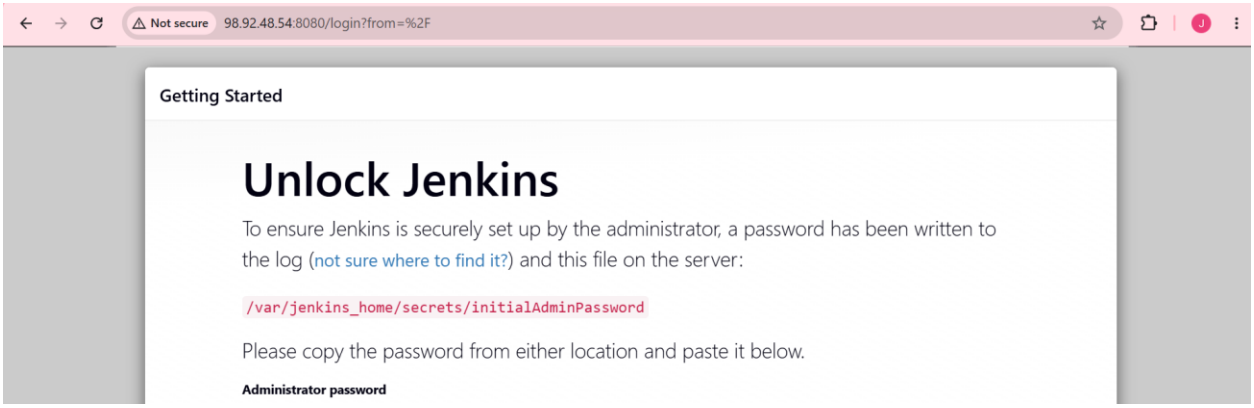
-p 8080:8080 — exposes the Jenkins web interface.

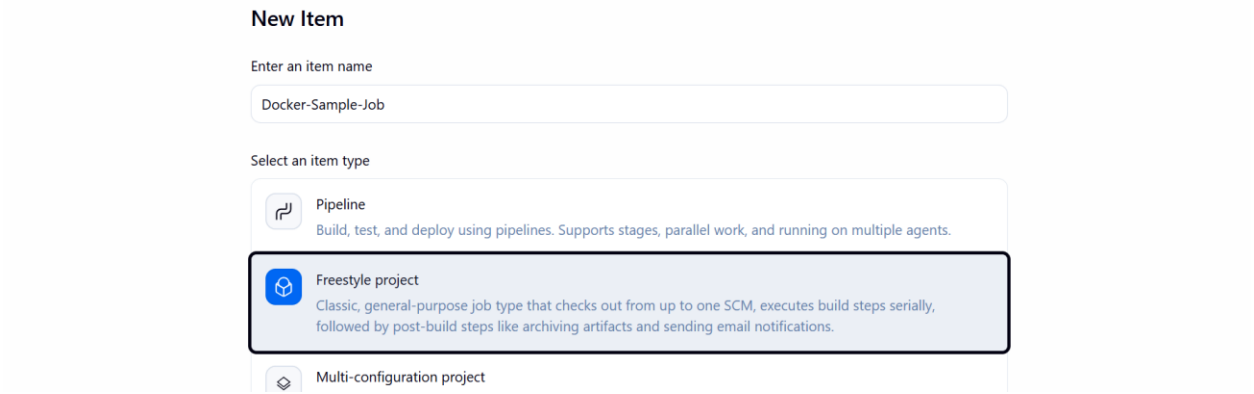
-p 50000:50000 — exposes the inbound agent port.

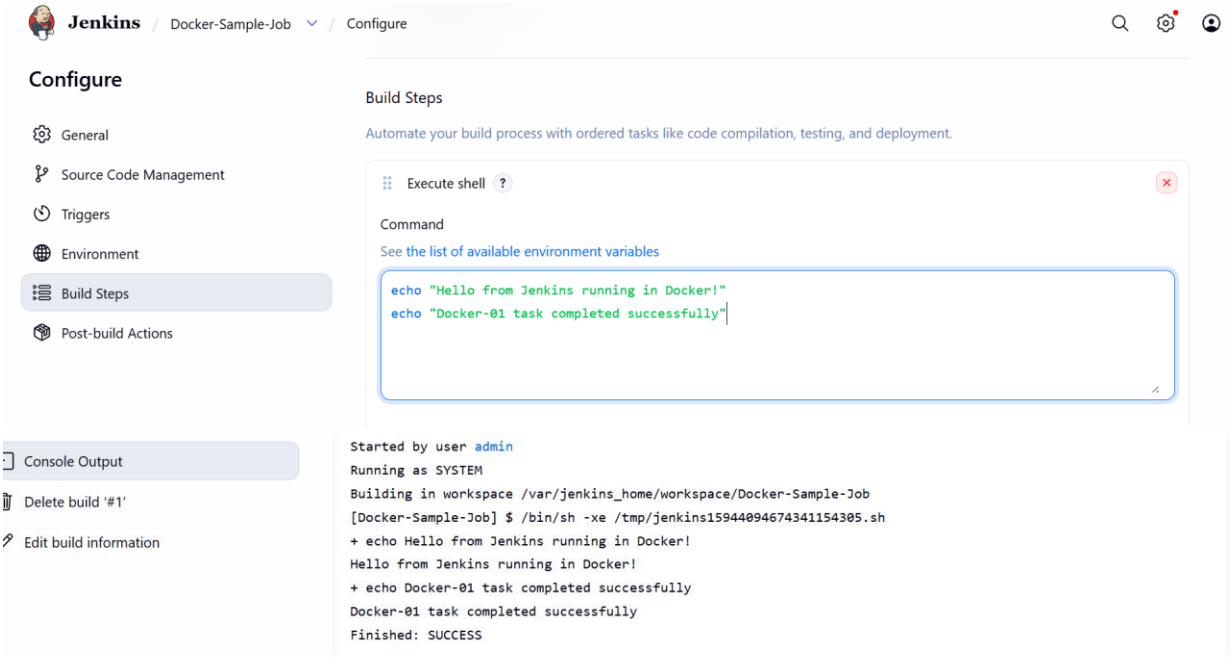




Add inbound rules allowing 8080






Jenkins / Docker-Sample-Job / Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Build Steps
Automate your build process with ordered tasks like code compilation, testing, and deployment.
Execute shell
Command
See the list of available environment variables
echo "Hello from Jenkins running in Docker!"
echo "Docker-01 task completed successfully"

Console Output
Delete build '#1'
Edit build information
Started by user admin
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/Docker-Sample-Job
[Docker-Sample-Job] $ /bin/sh -xe /tmp/jenkins15944094674341154305.sh
+ echo Hello from Jenkins running in Docker!
Hello from Jenkins running in Docker!
+ echo Docker-01 task completed successfully
Docker-01 task completed successfully
Finished: SUCCESS